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# Leveraging dynamic feature fusion of self & cross-attention for robust multimodal emotion recognition

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ABSTRACT

Recent multimodal emotion recognition (MER) approaches often assume that fusing audio and video modalities through cross-attention inherently yields superior feature representations for MER. However, our analysis reveals that different emotions manifest more prominently in specific modalities, leading to frequent misclassifications when an unsuitable modality dominates. Building on this, we propose a framework that dynamically adjusts each modality's contribution using weighted integration of self & cross-attention for emotion, called WISE-Net. In addition, as certain emotions benefit from merging complementary audio–video information, we introduce a cross-label emotion matching task to explicitly capture fine-grained emotion-related distinctions across modalities. Experiments confirm that WISE-Net surpasses state-of-the-art methods. Ablation studies highlight the significance of balancing self-attention for modality expressiveness and cross-attention for emotion-aware feature fusion.

## 1. Introduction

Emotion recognition technology can extend beyond conventional functional interfaces by enabling recognition systems to perceive user emotions and respond accordingly [1]. This capability facilitates natural human–computer interactions in domains such as in diverse domains such as mental health management, psychological therapy, and personalized education [2].

In addition, multimodal emotion recognition (MER) addresses the limitations of unimodal methods [3], which struggle with irony, background noise, and lighting conditions. By integrating text, speech, and vision, MER enables a comprehensive analysis of human emotional signals. This approach has been actively explored assuming different modalities offer complementary cues that together improve emotion recognition [4].

Modality fusion plays a crucial role in MER, with early and late fusion being the primary approaches [5]. While early fusion may introduce redundancy and increase training complexity, and late fusion may fail to capture cross-modal interactions, both approaches can lead to suboptimal integration [6]. To tackle these limitations, alternative fusion methods have emerged [7], with recent advancements focusing on cross-attention-based strategies [8]. Cross-attention strengthens inter-modal interactions by highlighting salient features in each modality. This allows models to integrate complementary information, and capture complex dependencies [9].

Meanwhile, each modality varies in effectiveness for recognizing emotion, as specific modalities more effectively capture certain emotions. For instance, angry emotions are often better conveyed through the audio than the video, as they are characterized by strong intonation and high-pitched variations. Similarly, neutral emotions may be harder to distinguish visually, as they lack distinct facial expressions or dynamic movements. Conversely, happy emotions tend to be more effectively detected via visual features, such as facial expressions and lip movements. Despite these distinctions, in-depth studies on the varying contributions of each modality for recognition remain insufficient.

As a first step, we conducted an unimodality experiment (Fig. 1) on the IEMOCAP benchmark to assess the relative importance of each modality in emotion recognition. In this experiment, we analyzed cases where the audio modality correctly recognized the emotion while the video modality misclassified it (Audio(O), Video(X)) and vice versa, where the video modality correctly identified the emotion while the audio modality failed (Audio(X), Video(O)). The results indicate that for neutral and angry emotions, the proportion of Audio(O), Video(X) was relatively high at 52.3% and 41.4%, respectively, while for happy and sad emotions, the proportion of Audio(X), Video(O) was relatively high at 33.2% and 29.4%, respectively. Fig. 1 (bottom) presents examples where the video modality misclassified emotions compared to the true labels. These examples show that relying solely on video can degrade performance. This underscores the need for self-attention

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| Emotion | Audio(X), Video(O) | Audio(O), Video(X) |
|---|---|---|
| Happy | **33.2%** | 7.9% |
| Neutral | 4.9% | **52.3%** |
| Sad | **29.4%** | 11.8% |
| Angry | 6.1% | **41.4%** |

**Fig. 1. Top**: The table presents the results of a unimodality experiment on the IEMOCAP benchmark, where the number of samples correctly classified by one modality (audio or video) but misclassified by the other is counted and reported as frequencies. Samples correctly classified by the audio modality but misclassified by the video modality are denoted as Audio(O), Video(X), while the reverse case is denoted as Audio(X), Video(O). In this experiment, we used WavLM [10] for audio and CLIP [11] for video.
**Bottom**: The misclassified video samples highlight discrepancies between the emotions recognized solely through the video modality and the actual emotion labels. In particular, among the samples labeled angry and neutral, many show ambiguity when recognized solely via video, often leading to misclassification, as shown in the examples.

before modality fusion to regulate modality-specific features and reduce the influence of misleading information, resulting in improved MER.

We propose WISE-Net (Fig. 2), a novel MER framework that integrates cross-attention and self-attention via learnable weights while incorporating an auxiliary task to enhance modality fusion. WISE-Net assigns a total of four learnable weights to cross-attention between audio and video and self-attention within each modality. This allows the model to leverage independent modality features through self-attention when fusion via cross-attention leads to ambiguous representations. This enables generalization across emotions, as the contribution of each modality to emotion recognition is not uniform across different emotional states.

Furthermore, to prevent incomplete modality fusion, we introduce the auxiliary task called cross-label emotion matching (CLEM). It utilizes the outputs of positive and negative pairs processed through cross-attention to reinforce emotion pair identification. CLEM explicitly learns the distinctions in negative pairs, where emotion labels between audio and video modalities do not align. By doing so, it strengthens the ability of cross-attention to more precisely model emotion distinctions. The effectiveness of the proposed mechanisms is demonstrated through experiments on multiple benchmark datasets, including IEMOCAP, CH-SIMS, and CH-SIMS v2. The core contributions of our study can be summarized as follows:

- To the best of our knowledge, we are the first to show that uniform fusion degrades MER performance when emotion cues are modality-specific. To address this, we propose a dynamic fusion strategy using self- and cross-attention with learnable weights for adaptive modality integration.
- We introduce cross-label emotion matching (CLEM) as an auxiliary module to reinforce fine-grained, emotion-distinctive representations throughout the fusion process.
- WISE-Net consistently outperforms state-of-the-art methods on diverse benchmark datasets, demonstrating strong generalization and robustness in multimodal emotion recognition. Ablation studies further validate the effectiveness of the proposed components in enhancing MER.

## 2. Proposed method

### 2.1. Learnable weighted sum for attention fusion

Equal contribution from all modalities can be suboptimal to emotion classification since the contribution of each modality to emotion varies. Considering this, we introduce a weighted sum module that prioritizes self-attention while dynamically adjusting modality importance. This approach optimizes modality fusion by assigning appropriate weights to each modality based on its relevance. This allows the model to learn optimal weights during training, emphasizing reliable cues and suppressing noise, as follows:

$$\begin{aligned} w_m &= \frac{\exp(w'_m)}{\sum_m \exp(w'_m)}, \\ F_{\text{fused}} &= w_{a2v} H_{A2V} + w_{v2a} H_{V2A} + w_a H_A + w_v H_V, \\ \hat{y} &= f_{\text{cls}}(F_{\text{fused}}), \end{aligned} \tag{1}$$

where $w_{a2v}, w_{v2a}, w_a, w_v$ are learnable weights for each modality interaction $m \in \{A2V, V2A, A, V\}$, which are automatically optimized during training and normalized via softmax to ensure a total sum of 1. Specifically, each $w_m$ is computed from an unnormalized scalar score $w'_m$, which is a learnable parameter associated with modality interaction $m$. $H_{A2V}, H_{V2A}, H_A, H_V$ denote the representations obtained from audio-to-video cross-attention, video-to-audio cross-attention, audio self-attention, and video self-attention, respectively. These representations are linearly combined using the normalized weights $w_m$ to obtain a comprehensive multimodal feature $F_{\text{fused}}$ which captures both modality-specific and cross-modal interactions. The final fused emotion representation $F_{\text{fused}}$ is obtained by applying the learnable weights to these emotion representation vectors. $f_{\text{cls}}$ denotes the emotion classification head.

The model trained with the weighted sum module has advantages. It first leverages self-attention to focus on the most relevant modality, enabling optimal fusion even in unimodal-dominant scenarios. Furthermore, unlike traditional concatenation-based fusion, the learnable weighting mechanism dynamically adjusts modality importance. It also improves robustness by allocating higher weights to more reliable inputs.

### 2.2. Cross-label emotion matching (CLEM)

CLEM is an auxiliary task designed to improve cross-attention by leveraging emotion mismatches between modalities as supervision. We construct positive and negative pairs within a given batch based on the alignment of emotion labels between the audio and video modalities. Positive pairs $(A_i, V_i)$ share the same emotion and originate from the same instance. Negative pairs $(A_i, V_j)$ are from different instances with different emotion labels. After constructing sample pairs, each pair was processed using audio-to-video ($a2v$) and video-to-audio ($v2a$) cross-attention mechanisms. For negative pairs, the model learns to explicitly distinguish emotional discrepancies through cross-attention. The model subsequently classifies the emotion pair based on the learned representations. The cross-entropy loss is applied to each modality pair ($a2v$, $v2a$), and the final loss function is defined as follows:

$$L_{\text{CLEM}} = \frac{1}{4} \sum_{m \in \{a2v, v2a\}} \left( \mathcal{L}_{CE}(\hat{y}_m^{\text{pos}}, y_{\text{aux}}) + \mathcal{L}_{CE}(\hat{y}_m^{\text{neg}}, y_{\text{aux}}) \right), \tag{2}$$

where $\mathcal{L}_{CE}(\hat{y}, y)$ represents the cross-entropy loss, $\hat{y}_m^{\text{pos}}$ and $\hat{y}_m^{\text{neg}}$ denote the predicted classes for positive and negative samples, respectively. The variable $m$ specifies the modality mapping direction, where $a2v$ cross-attention integrates audio features into video features, and $v2a$ cross-attention integrates video features into audio features. The emotion label of auxiliary task $y_{\text{aux}}$ is used exclusively in the CLEM task and is a numeric value assigned to each emotion pair $(y_a, y_v)$. The total number of auxiliary labels ($y_{\text{aux}}$) is 16, based on the fixed emotion order [happy, neutral, sad, angry] and enumerated in row-major fashion.

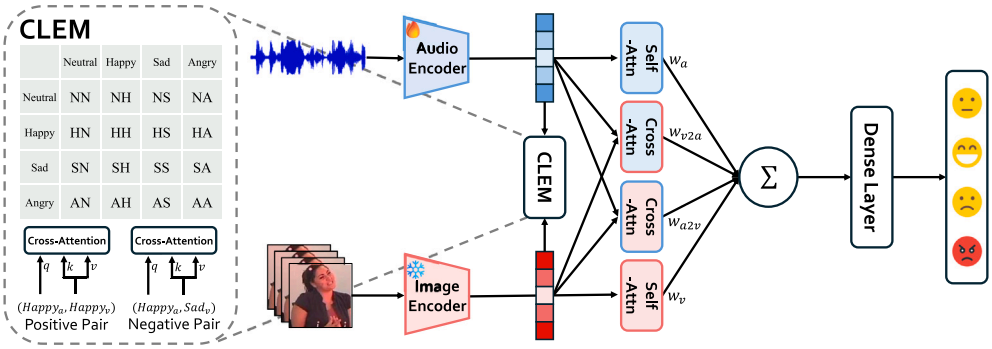


**Fig. 2.** Overall architecture of the proposed framework, WISE-Net. The audio encoder, WavLM, and the image encoder, CLIP, extract features from their respective modalities. The audio encoder is used with trainable parameters, while the image encoder is kept frozen during training. Since the audio features (e.g., $H_{A2V}, H_A$) are 1024-dimensional and the video features (e.g., $H_{V2A}, H_V$) are 768-dimensional, we project all features to 768 dimensions using learnable linear layers before fusion. The fused representation is then passed through a final dense layer for emotion classification.

**Algorithm 1:** Learnable Weighted Sum for Attention Fusion

**Input:**
Cross-attention representations $H_{\text{A2V}}, H_{\text{V2A}}$
Self-attention representations $H_A, H_V$
Learnable weights $w_{\text{A2V}}, w_{\text{V2A}}, w_A, w_V$
Emotion classification head $f_{\text{cls}}$
**Output:**
Final emotion prediction $\hat{y}$

**Normalize Weights:**
$w_m \leftarrow \frac{\exp(w'_m)}{\sum_m \exp(w'_m)}$, where $m \in \{\text{A2V, V2A, A, V}\}$
**Compute Weighted Sum:**
$F_{\text{fused}} \leftarrow w_{\text{A2V}} H_{\text{A2V}} + w_{\text{V2A}} H_{\text{V2A}} + w_A H_A + w_V H_V$
**Compute Final Prediction:**
$\hat{y} \leftarrow f_{\text{cls}}(F_{\text{fused}})$
**Compute Each (Fusion and CLEM) Loss:**
$L_{\text{fusion}} \leftarrow \mathcal{L}_{\text{CE}}(\hat{y}, y)$
$L_{\text{CLEM}} \leftarrow$ from Algorithm 2
**Compute Total Loss:**
$L_{\text{total}} = \lambda L_{\text{CLEM}} + L_{\text{fusion}}$, where $\lambda = 0.0001$

**Algorithm 2:** $L_{\text{CLEM}}$: Cross-Label Emotion Matching (CLEM)

**Input:**
Audio and video feature embeddings $A$, $V$
Emotion labels $y_A$, $y_V$
Auxiliary emotion-pair labels $y_{\text{aux}}$
Cross-attention layers $f_{a2v}$, $f_{v2a}$
Auxiliary classification head $f_{\text{aux}}$
**Output:**
Auxiliary CLEM loss $L_{\text{CLEM}}$

**Construct Positive and Negative Pairs:**
$P_{\text{pos}} \leftarrow \{(A_i, V_i) \mid y_{A,i} = y_{V,i}\}$
$P_{\text{neg}} \leftarrow \{(A_i, V_j) \mid y_{A,i} \neq y_{V,j}\}$
**Compute Prediction for Positive Pairs:**
**foreach** $(A_i, V_i) \in P_{pos}$ **do**
  $\hat{y}^{\text{pos}}_{A2V} \leftarrow f_{\text{aux}}(f_{a2v}(A_i, V_i))$
  $\hat{y}^{\text{pos}}_{V2A} \leftarrow f_{\text{aux}}(f_{v2a}(V_i, A_i))$
**Compute Prediction for Negative Pairs:**
**foreach** $(A_i, V_j) \in P_{neg}$ **do**
  $\hat{y}^{\text{neg}}_{A2V} \leftarrow f_{\text{aux}}(f_{a2v}(A_i, V_j))$
  $\hat{y}^{\text{neg}}_{V2A} \leftarrow f_{\text{aux}}(f_{v2a}(V_j, A_i))$
**Compute Auxiliary (CLEM) Loss:**
$L_{\text{CLEM}} = \frac{1}{4} \sum_{m \in \{\text{A2V,V2A}\}} \left(\mathcal{L}_{\text{CE}}(\hat{y}^{\text{pos}}_m, y_{\text{aux}}) + \mathcal{L}_{\text{CE}}(\hat{y}^{\text{neg}}_m, y_{\text{aux}})\right)$

Positive and negative pairs are assigned distinct auxiliary labels depending on the specific emotion combination. This design enables the CLEM task to explicitly leverage the mismatch between audio and video emotional cues, encouraging the model to learn discriminative features that capture subtle cross-modal discrepancies. By supervising these mismatched pairs with unique labels, the model is guided to recognize the nuanced differences in emotional expression, thereby enhancing its sensitivity to emotion recognition even in modality samples with mismatched emotional cues.

CLEM offers several key benefits. First, by leveraging both positive and negative pairs, the model learns richer emotion representations, mitigating over-reliance on a single modality. Second, by capturing diverse emotional patterns, CLEM enhances domain adaptability, enabling the model to handle previously unseen emotion combinations.

## 3. Experiments and results

### 3.1. Experimental setup

**DATASET** IEMOCAP [12] is a widely used multimodal benchmark for emotion recognition, consisting of 7,929 utterances annotated with four emotion categories: happy (including excited), neutral, sad, and angry. The dataset comprises five sessions involving ten speakers (five male and five female) and provides aligned speech, video, and text modalities. Due to its high-quality scripted and improvised dialogs, it enables fine-grained analysis of affective expressions in controlled dyadic interactions. CH-SIMS [13] is a Chinese multimodal sentiment analysis dataset containing 2,281 video samples with aligned text, audio, and visual modalities. We formulate the task as a 3-class classification problem by dividing the averaged sentiment scores into negative, neutral, and positive using thresholds at −0.5 and 0.5. CH-SIMS v2.0 [14] is an enhanced version of CH-SIMS that improves the consistency and diversity of emotion expression across modalities while retaining the original 2,281 samples. A key innovation in this version is the introduction of AV-Mixup, a multimodal data augmentation technique that mixes audio and visual modalities from different samples to create training instances with more diverse and uncertain emotion patterns. This encourages the model to better generalize to inconsistent or ambiguous affective cues. The annotation scheme remains the same as CH-SIMS, and we apply the same 3-class classification setting in our experiments.

**Experimental details** We used WavLM [10] and CLIP [11] as the audio and image encoders with output dimensions of 1024 and 768, respectively. Training was performed with a batch size of 16 and a learning rate of 3e-5 using the AdamW optimizer. Audio inputs were

**Table 1**
Comparison of state-of-the-art MER methods in terms of unweighted accuracy (UA) and weighted accuracy (WA).

| Method | Modality | Year | UA (%) | WA (%) |
|---|---|---|---|---|
| FTSLLM [15] | A+T | 2023 | 78.50 | 77.70 |
| MSMSER [16] | A+T | 2023 | 76.40 | 75.20 |
| ATMF [17] | A+T | 2023 | 79.71 | 78.42 |
| RMER [18] | A+T | 2023 | 77.00 | 76.00 |
| MTTX [19] | A+T | 2023 | 75.00 | 74.50 |
| CRNN-SAN [20] | A+T | 2023 | 79.95 | 78.82 |
| BAM [21] | A+T | 2023 | 77.00 | 75.50 |
| MSER [8] | A+T | 2024 | 76.56 | 77.20 |
| MMAN [22] | A+V | 2020 | – | 73.94 |
| AM-FBP [23] | A+V | 2021 | 75.49 | – |
| MCWSA-CMHA [24] | A+V | 2022 | 78.90 | – |
| GCNet [25] | A+V | 2023 | 78.36 | – |
| Foal-Net [26] | A+V | 2024 | 80.10 | 79.45 |
| **WISE-Net** | A+V | 2025 | **81.22** | **80.37** |

6-second clips at 16 kHz, and video inputs consisted of 180 frames. Since recordings involved two speakers, we applied frame-wise preprocessing to separate them. Given the limited size of the datasets, we carefully designed our training strategy to prevent overfitting and enhance the robustness of WISE-Net. In particular, we applied dropout to the classification layers and relied on the optimizer's built-in regularization. These regularization techniques helped promote generalization, as evidenced by the model's consistent performance across IEMOCAP, CH-SIMS, and CH-SIMS v2 (Table 1, Table 3).

**Evaluation Metrics** Unweighted Accuracy (UA) is computed as the average of recall scores across all classes, giving equal importance to each class regardless of its frequency. In contrast, Weighted Accuracy (WA) is the overall classification accuracy, reflecting the true distribution of classes in the dataset. For all datasets, UA, WA, and F1-score are used as the primary evaluation metrics.

### 3.2. Performance comparison with existing methods

As shown in Table 1, WISE-Net outperforms all prior approaches, attaining UA of 81.22% and WA of 80.37%, outperforming the latest audio–video (A+V) based model, Foal-Net [26] (UA: 80.10%, WA: 79.45%). These results confirm that the proposed model, WISE-Net, effectively captures the relationships between audio and video modalities.

### 3.3. Ablation study

Table 2 presents the contribution of each module in WISE-Net. Self-attention alone (w/o Cross-Attn) outperforms cross-attention alone (w/o Self-Attn), emphasizing the importance of modeling intra-modality emotional features. capturing fine-grained intra-modal patterns is essential for recognizing nuanced emotions. The complete model, which combines both mechanisms, achieves the highest overall performance, demonstrating their complementary effects.

As shown in Table 3, WISE-Net consistently exceeds the baseline Foal-Net across both CH-SIMS and CH-SIMS v2, yielding higher UA, WA, and notably improved F1-scores for the neutral class. This result highlights WISE-Net's strength in handling emotionally ambiguous categories under multi-domain conditions. We focus only on the neutral class when reporting these ablation results, as our goal is to assess the specific effect of CLEM on emotionally ambiguous expressions.

On the IEMOCAP benchmark, the F1-score for the neutral class remains relatively low. To investigate, we examine how the CLEM mechanism interacts with ambiguous expressions. We posit that CLEM improves class separability by learning sharper decision boundaries. While this benefits the discrimination of distinct emotions, it can hinder classification of borderline cases like neutral. In such cases, CLEM may push the representation toward more confident emotion classes, reducing sensitivity to emotional ambiguity.

**Table 2**
Ablation study results on the IEMOCAP benchmark. The table presents F1-scores (%) for each emotion category: Happy, Neutral, Sad, and Angry, and UA and WA for the overall four emotions.

| | Happy | Neutral | Sad | Angry | All | |
|---|---|---|---|---|---|---|
| | | F1-scores(%) | | | UA | WA |
| w/o Self-Attn | 82.64 | 73.18 | 78.30 | 74.24 | 79.07 | 78.07 |
| w/o Cross-Attn | 83.30 | 74.12 | 79.38 | 75.42 | 79.69 | 78.75 |
| w/o Weighted Sum | 84.34 | 76.68 | 81.94 | 77.08 | 80.65 | 80.54 |
| w/o CLEM | 84.31 | **77.00** | 82.56 | 77.14 | 80.94 | **80.75** |
| **WISE-Net** | **84.86** | 74.70 | **83.14** | **78.32** | **81.22** | 80.37 |

**Table 3**
Comparison of WISE-Net and the baseline (Foal-Net) on CH-SIMS and CH-SIMS v2. Reported metrics include the neutral-class F1-score and overall UA/WA, highlighting both fine-grained and general performance differences. $\lambda = 0.001$.

| Benchmark | Model | Neutral | UA | WA |
|---|---|---|---|---|
| CH-SIMS | Baseline | 60.22 | 63.86 | 63.68 |
| | w/o CLEM | 60.00 | **64.98** | 63.68 |
| | WISE-Net | **61.25** | 64.08 | **64.11** |
| CH-SIMS v2 | Baseline | 72.55 | 65.02 | 68.28 |
| | w/o CLEM | 68.63 | 69.58 | 67.94 |
| | **WISE-Net** | **69.52** | **69.90** | **68.46** |

To validate this, we exclude negative pairs involving neutral samples during CLEM training. This improves the neutral F1-score, highlighting a trade-off: although CLEM enhances class separation and overall discriminability, it may reduce performance on subtle categories like neutral.

## 4. Conclusions

We found that self-attention plays a crucial role in emotion recognition and that its combination with cross-attention enables more effective multimodal fusion. Building on this, we propose WISE-Net, a novel MER framework. With the CLEM auxiliary task, WISE-Net enhances cross-attention while preserving modality-specific features through self-attention, addressing key limitations of existing fusion methods. By balancing inter- and intra-modal learning, it achieves robust performance, outperforming state-of-the-art models in UA and WA across multiple datasets and emotion categories.

## CRediT authorship contribution statement

**Jian Kim:** Writing – original draft, Validation, Methodology. **Hyunoh Kang:** Writing – review & editing, Visualization. **Haneul Pyeon:** Writing – review & editing, Visualization. **Dahuin Jung:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization.

## Declaration of competing interest

The authors declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## References


[1] C.-M. Chen, H.-P. Wang, Using emotion recognition technology to assess the effects of different multimedia materials on learning emotion and performance, Libr. Inf. Sci. Res. 33 (3) (2011) 244–255.

[2] S. Graham, C. Depp, E.E. Lee, C. Nebeker, X. Tu, H.-C. Kim, D.V. Jeste, Artificial intelligence for mental health and mental illnesses: an overview, Curr. Psychiatry Rep. 21 (2019) 1–18.

[3] R.A. Khalil, E. Jones, M.I. Babar, T. Jan, M.H. Zafar, T. Alhussain, Speech emotion recognition using deep learning techniques: A review, IEEE Access 7 (2019) 117327–117345.

[4] X. Gu, Y. Shen, J. Xu, Multimodal emotion recognition in deep learning: a survey, in: 2021 International Conference on Culture-Oriented Science & Technology, ICCST, IEEE, 2021, pp. 77–82.

[5] H. Gunes, M. Piccardi, Affect recognition from face and body: early fusion vs. late fusion, in: 2005 IEEE International Conference on Systems, Man and Cybernetics, vol. 4, IEEE, 2005, pp. 3437–3443.

[6] S. Mai, Y. Zeng, H. Hu, Multimodal information bottleneck: Learning minimal sufficient unimodal and multimodal representations, IEEE Trans. Multimed. 25 (2022) 4121–4134.

[7] J. Ngiam, A. Khosla, M. Kim, J. Nam, H. Lee, A.Y. Ng, et al., Multimodal deep learning., in: ICML, vol. 11, 2011, pp. 689–696.

[8] M. Khan, W. Gueaieb, A. El Saddik, S. Kwon, MSER: Multimodal speech emotion recognition using cross-attention with deep fusion, Expert Syst. Appl. 245 (2024) 122946.

[9] H. Lee, S. Suniljit, Y.S. Ong, Dynamic multimodal sentiment analysis: Leveraging cross-modal attention for enabled classification, 2025, arXiv preprint arXiv:2501.08085.

[10] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao, et al., Wavlm: Large-scale self-supervised pre-training for full stack speech processing, IEEE J. Sel. Top. Signal Process. 16 (6) (2022) 1505–1518.

[11] A. Radford, J.W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., Learning transferable visual models from natural language supervision, in: International Conference on Machine Learning, PMLR, 2021, pp. 8748–8763.

[12] C. Busso, M. Bulut, C.-C. Lee, A. Kazemzadeh, E. Mower, S. Kim, J.N. Chang, S. Lee, S.S. Narayanan, IEMOCAP: Interactive emotional dyadic motion capture database, Lang. Resour. Eval. 42 (2008) 335–359.

[13] W. Yu, H. Xu, F. Meng, Y. Zhu, Y. Ma, J. Wu, J. Zou, K. Yang, Ch-sims: A chinese multimodal sentiment analysis dataset with fine-grained annotation of modality, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 2020, pp. 3718–3727.

[14] Y. Liu, Z. Yuan, H. Mao, Z. Liang, W. Yang, Y. Qiu, T. Cheng, X. Li, H. Xu, K. Gao, Make acoustic and visual cues matter: Ch-sims v2. 0 dataset and av-mixup consistent module, in: Proceedings of the 2022 International Conference on Multimodal Interaction, 2022, pp. 247–258.

[15] W. Yang, S. Fukayama, P. Heracleous, J. Ogata, Exploiting fine-tuning of self-supervised learning models for improving bi-modal sentiment analysis and emotion recognition, in: INTERSPEECH, 2022, pp. 1998–2002.

[16] S. Wang, Y. Ma, Y. Ding, Exploring complementary features in multi-modal speech emotion recognition, in: ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, IEEE, 2023, pp. 1–5.

[17] D. Sun, Y. He, J. Han, Using auxiliary tasks in multimodal fusion of wav2vec 2.0 and bert for multimodal emotion recognition, in: ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, IEEE, 2023, pp. 1–5.

[18] B. Lin, L. Wang, Robust multi-modal speech emotion recognition with ASR error adaptation, in: ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, IEEE, 2023, pp. 1–5.

[19] J. He, M. Wu, M. Li, X. Zhu, F. Ye, Multilevel transformer for multimodal emotion recognition, in: ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, 2023, pp. 1–5.

[20] B. Maji, M. Swain, R. Guha, A. Routray, Multimodal emotion recognition based on deep temporal features using cross-modal transformer and self-attention, in: ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, IEEE, 2023, pp. 1–5.

[21] Z. Zhao, Y. Wang, Y. Wang, Knowledge-aware Bayesian co-attention for multi-modal emotion recognition, in: ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, IEEE, 2023, pp. 1–5.

[22] Z. Pan, Z. Luo, J. Yang, H. Li, Multi-modal attention for speech emotion recognition, in: Proc. Interspeech 2020, 2020, pp. 364–368, http://dx.doi.org/10.21437/Interspeech.2020-1653.

[23] H. Zhou, J. Du, Y. Zhang, Q. Wang, Q.-F. Liu, C.-H. Lee, Information fusion in attention networks using adaptive and multi-level factorized bilinear pooling for audio-visual emotion recognition, IEEE/ACM Trans. Audio Speech Lang. Process. 29 (2021) 2617–2629.

[24] J. Zheng, S. Zhang, Z. Wang, X. Wang, Z. Zeng, Multi-channel weight-sharing autoencoder based on cascade multi-head attention for multimodal emotion recognition, IEEE Trans. Multimed. (2022).

[25] Z. Lian, L. Chen, L. Sun, B. Liu, J. Tao, GCNet: graph completion network for incomplete multimodal learning in conversation, IEEE Trans. Pattern Anal. Mach. Intell. (2023).

[26] Q. Li, Y. Gao, Y. Wen, C. Wang, Y. Li, Enhancing modal fusion by alignment and label matching for multimodal emotion recognition, 2024, arXiv preprint arXiv:2408.09438.